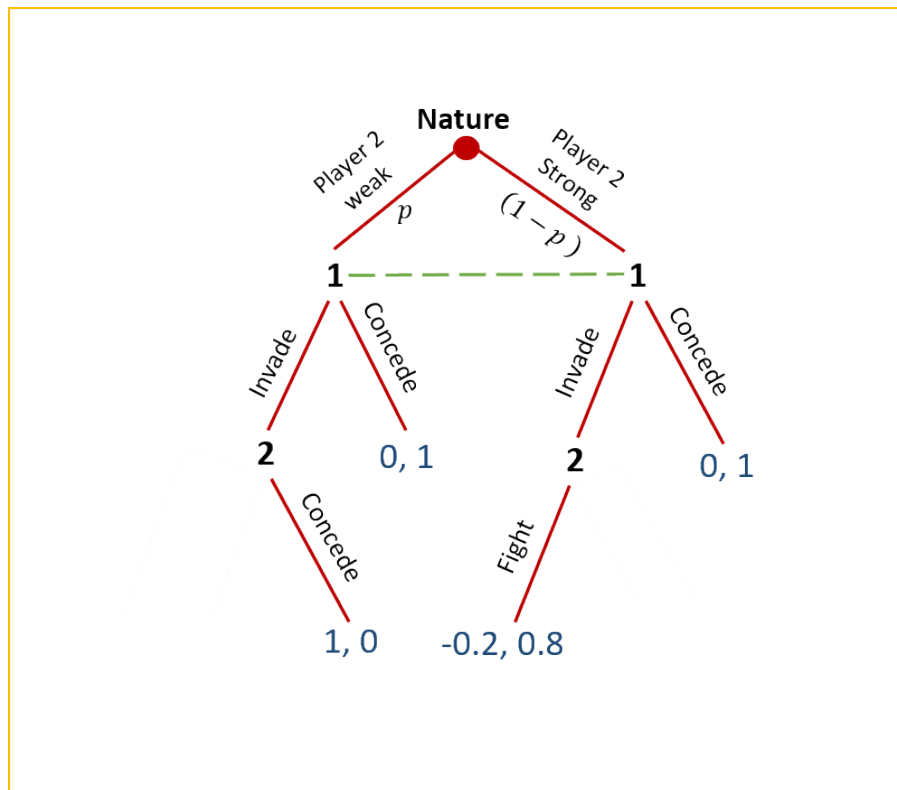




Now, how do we solve for the game's analytical equilibrium?¹²

Here, Player 2 applies backward induction to find what's known as a *Perfect Bayesian Equilibrium* (PBE). As we already know, if Player 2 is the weak type and Player 1 has chosen to invade, then Player 2 should concede. If he is the strong type, then Player 2 should fight. We also know that Player 1 recognizes that she gets a payoff of \$0 if she concedes in the first round, regardless of Player 2's type. If she instead chooses to invade in the first round, then Player 1's expected payoff from invading is $p - 0.2(1 - p) = 1.2p - 0.2$. This is merely the weighted average of Player 1's expected payoff when Player 2 is weak and her expected payoff when Player 2 is strong. Thus, invade is a better strategy than concede for Player 1 when $1.2p - 0.2 > 0 \implies p > 1/6$. In other words, if the probability that Player 1 assigns to Player 2 being weak is greater than one-sixth, Player 1 should choose to invade in the first round. Otherwise, Player 1 should concede and be done with it.

What's the outcome when you and your classmates play this more complicated version of the Escalation Game?

BURNING BRIDGES GAME

This game shares starkly similar features with the Escalation Game, but there is no uncertainty (thus, the analytical equilibrium is an SPE rather than a PBE). The SPE has much to say about the relationship between two tenacious competitors. Spaniel (2011) portrays the game as follows:

12. This equilibrium is known as a *Perfect Bayesian Equilibrium* (PBE) rather than an SPE because of the uncertainty that at least one of the players is forced to contend with. Similar to Nash, Thomas Bayes is considered a towering figure. He was an 18th-century English statistician, philosopher, and Presbyterian minister who is known for formulating a specific case of the theorem that bears his name: Bayes Theorem. Bayes never published his theory himself—his notes were edited and published posthumously.